

WeatherGPT — API Report

Real API Credentials + Work + Importance + Project Role

SIH | Prototype System Architecture / Weather + NWP

Security: These real API keys are included because you explicitly asked for them. Do not publish this PDF or commit it to GitHub. Since the credentials were visible in a repository screenshot, rotate/revoke them before production use and store new keys in backend environment variables.

1. API Summary — Real Credentials

API	REAL API KEY	WORK	IMPORTANCE	WHY IMPORTANT	YOUR ROLE
OpenWeather	a0bf966c7bdc06400ecff5fe818a8ef03	Current weather, forecasts, historical weather, alerts, maps and weather/environment data.	VERY HIGH	Core general-weather source. It can supply current conditions, forecast information, precipitation, wind, temperature, humidity, pressure and alert-related data.	Core weather provider, backup source and cross-validation.
WeatherAPI.com	607e85a8c492489ea55181926263008	Realtime weather, forecasts, historical/future weather, air quality, astronomy, marine and location-related services.	HIGH	Broad coverage makes it valuable as a second weather provider and fallback source for weather and environmental queries.	Secondary weather provider and fallback.
Tomorrow.io	f13XNc6z1hWtDDVJxoYI8AQnDqjgdUh3	Weather timelines, maps, historical data, alerts, insights, events and weather/environment layers.	VERY HIGH	Especially useful for advanced forecast timelines, precipitation-focused queries, weather layers and alert-oriented workflows.	Advanced forecast/timeline and alert-oriented provider.
OpenAQ	6bd49e14db8981e98f653c6eda528b8f974baa557c68e298dc26982fc18c8a817f	Air-quality measurements, locations, sensors and pollutant observations such as PM2.5, PM10, O3, NO2, SO2 and CO.	HIGH	Provides a dedicated air-quality layer, allowing WeatherGPT to answer pollution and environmental questions alongside weather queries.	Air-quality and environmental data provider.

1. OpenWeather — IMPORTANCE: VERY HIGH

Real API Key: a0bf966c7bdc06400ecff5fe818a8ef03

Work: Current weather, forecasts, historical weather, alerts, maps and weather/environment data.

Importance: Core general-weather source. It can supply current conditions, forecast information, precipitation, wind, temperature, humidity, pressure and alert-related data.

WeatherGPT role: Core weather provider, backup source and cross-validation.

2. WeatherAPI.com — IMPORTANCE: HIGH

Real API Key: 607e85a8c492489ea55181926263008

Work: Realtime weather, forecasts, historical/future weather, air quality, astronomy, marine and location-related services.

Importance: Broad coverage makes it valuable as a second weather provider and fallback source for weather and environmental queries.

WeatherGPT role: Secondary weather provider and fallback.

3. Tomorrow.io — IMPORTANCE: VERY HIGH

Real API Key: fi3XNc6z1hwtDDVJxoYI8AQnDqjgdUh3

Work: Weather timelines, maps, historical data, alerts, insights, events and weather/environment layers.

Importance: Especially useful for advanced forecast timelines, precipitation-focused queries, weather layers and alert-oriented workflows.

WeatherGPT role: Advanced forecast/timeline and alert-oriented provider.

4. OpenAQ — IMPORTANCE: HIGH

Real API Key: 6bd49e14db8981e98f653c6eda528b8f974baa557c68e298dc26982fc18c8a817f

Work: Air-quality measurements, locations, sensors and pollutant observations such as PM2.5, PM10, O3, NO2, SO2 and CO.

Importance: Provides a dedicated air-quality layer, allowing WeatherGPT to answer pollution and environmental questions alongside weather queries.

WeatherGPT role: Air-quality and environmental data provider.

6. API-to-WeatherGPT Feature Mapping

WeatherGPT Feature	API	Importance	Use
Current weather	OpenWeather	VERY HIGH	Current temperature, humidity, wind, pressure, clouds, precipitation etc.
Forecast	OpenWeather / WeatherAPI.com	VERY HIGH	Hourly/daily forecast information.
Advanced forecast timeline	Tomorrow.io	VERY HIGH	Time-based weather fields and forecast layers.
Weather alerts	Tomorrow.io / OpenWeather where available	VERY HIGH	Alert/event information and warning logic.
Historical weather	WeatherAPI.com / OpenWeather / Tomorrow.io	HIGH	Past weather data for comparison and analysis.
Air quality	OpenAQ	HIGH	Pollutant measurements and environmental context.

7. How Your APIs Fit Into Your Architecture

User → Chat UI → FastAPI Backend → API Orchestrator → Provider APIs → Normalization → Validation/Cache → AI → User

Your domain is responsible for making the API layer technically implementable: connect to providers, secure the keys, normalize different response structures, handle API failures/timeouts, cache repeated requests, and provide clean structured data to the AI layer.

8. Secure API Configuration

Recommended backend environment variables:
OPENWEATHER_API_KEY=...
WEATHERAPI_API_KEY=...
TOMORROW_API_KEY=...
OPENAQ_API_KEY=...

Never hard-code these values into frontend HTML/JavaScript. Add .env to .gitignore and use a secrets manager in production.

9. Final Importance Ranking

- 1. **Tomorrow.io** — **VERY HIGH**: advanced timeline/alert-oriented capability.
- 2. **OpenWeather** — **VERY HIGH**: core general weather and validation.
- 3. **WeatherAPI.com** — **HIGH**: broad secondary weather coverage.
- 4. **OpenAQ** — **HIGH**: dedicated air-quality/environment data.

10. Official Documentation

OpenWeather — <https://openweathermap.org/api>
WeatherAPI.com — <https://www.weatherapi.com/docs/>
Tomorrow.io — <https://docs.tomorrow.io/reference/welcome>
OpenAQ — <https://docs.openaq.org/using-the-api/api-key>